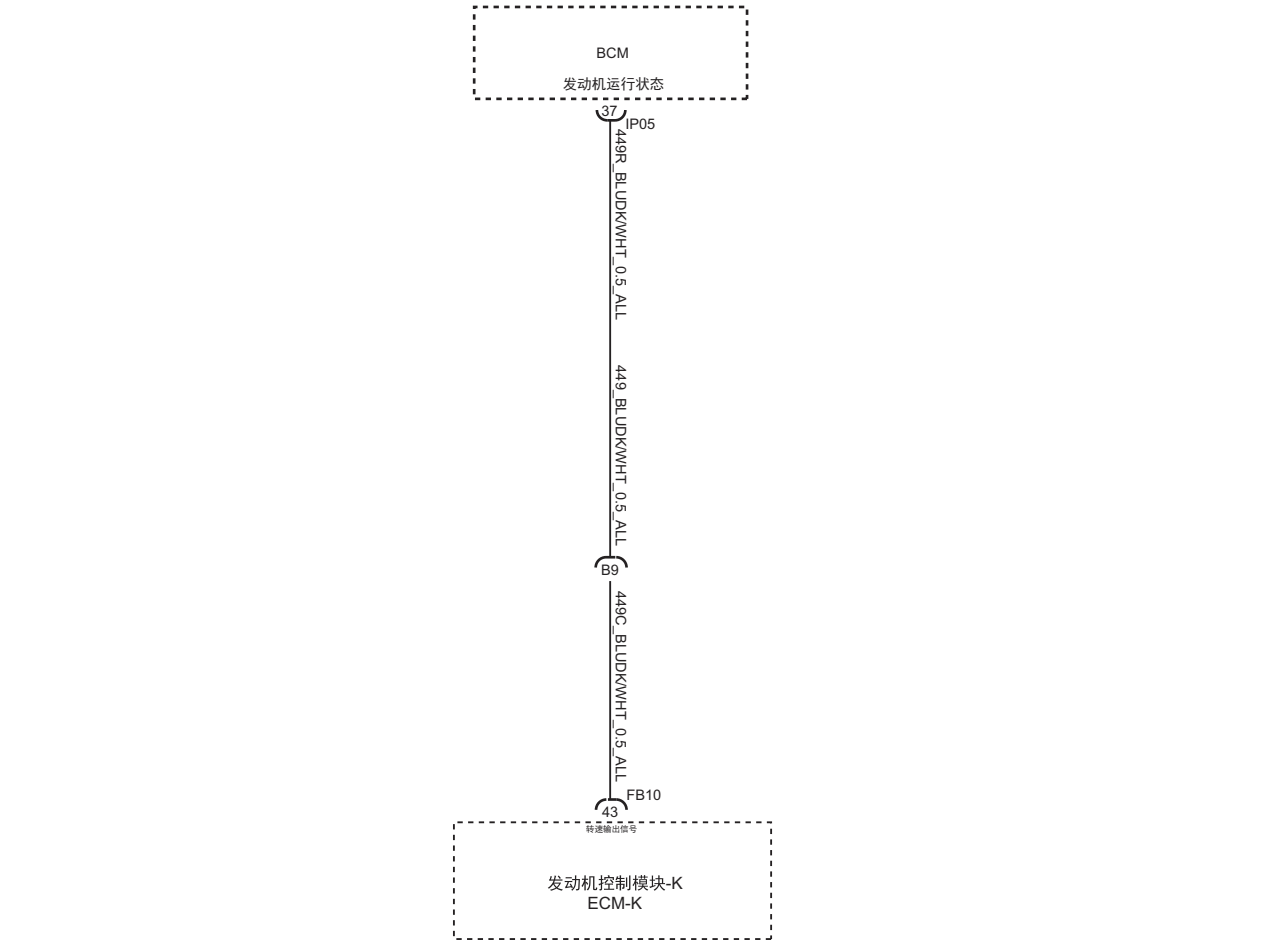
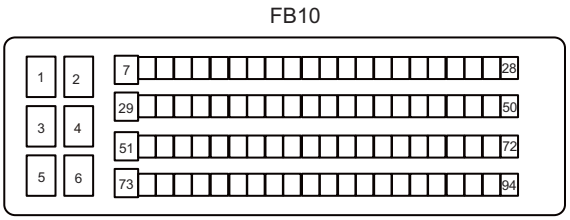
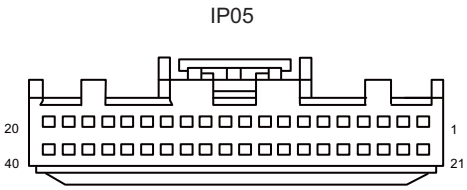
P0725

Description of fault diagnosis codes
P0725-11：Short circuit to ground error on the engine speed output
P0725-12：Short circuit to battery error on the engine speed output
P0725-13：No load error on the engine speed output
P0725-4B：Over Temperature error on the engine speed output
Detection tools
1 Multimeter and diagnostic apparatus 2 Oscilloscope
Possible causes
• Fault in harness or harness plug • Damages of engine control module
Description of circuit diagram
Engine control module detects the failure of speed output drive circuit.

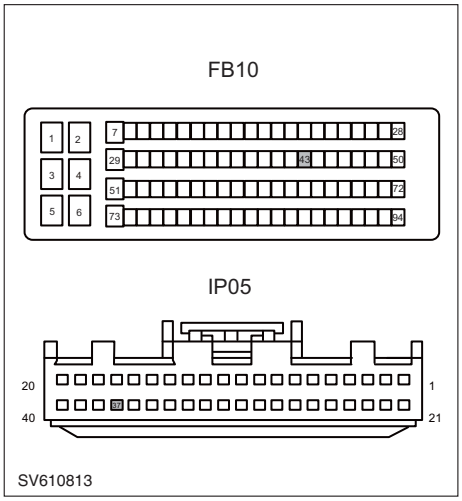
Circuit diagram



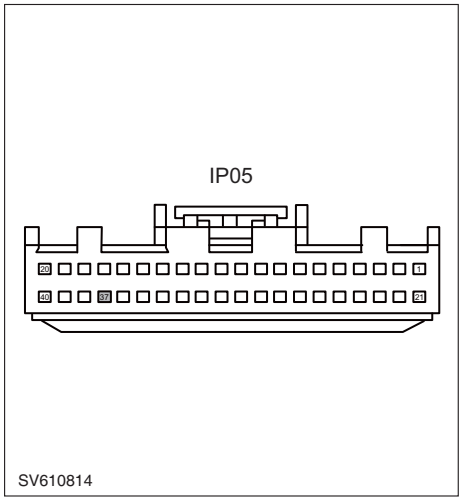




SV610807

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check if there is any damage, bad contact, aging and looseness phenomenon on harness plug of engine control module and body control module. 4 Check whether the result is normal?
		No ➞ Repair or replace the faulted parts.
Yes ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the engine control system with diagnostic apparatus. 4 If there is any fault code except for P0725?
		Yes ➞ Refer to: DTC Summary List (EU5).
No ↓		
Step 3	Check revolving speed output signal circuit (open circuit).	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug FB10 on engine control module. 3 Disconnect BCM harness plug IP05. 4 Measure the resistance between harness terminal on engine control module and BCM harness terminal. Measuring circuit: resistance between terminal 43 of plug FB10 and terminal 37 of plug IP05. **Standard value: ≈ 0Ω** 5 Check whether the result is normal?
		No ➞ Repair or replace the faulted parts.
Yes ↓		

Step 4	Check revolving speed output signal circuit (short circuit).



1 Turn the ignition switch to OFF.

2 Disconnect the harness plug of engine control module.

3 Disconnect BCM harness plug.

4 Measure the circuit between BCM harness terminal and ground.

Measuring circuit: resistance between terminal 37 of plug IP05 and ground.

Standard value: ∞

Measuring circuit: voltage between terminal 37 of plug IP05 and ground.

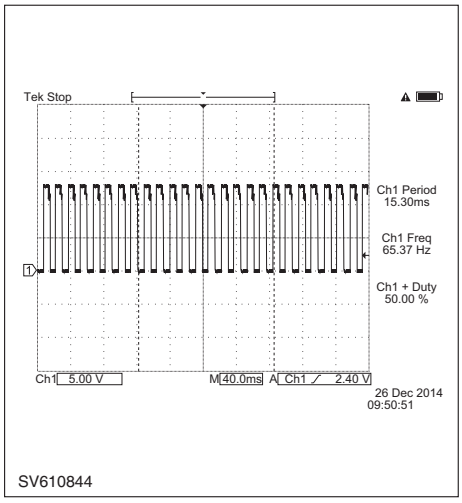
Standard value: ≈ 0V

5 Check whether the result is normal?

No → Repair or replace the faulted parts.

Yes ↓

Step 5	Check the revolving speed output signal waveform of engine.



1 Turn the ignition switch to OFF.

2 Connect the harness plug of engine control module.

3 Connect BCM harness plug.

4 Turn the ignition switch to ON position.

5 Start the engine.

6 Measure the waveform of harness plug terminal on engine control module with oscilloscope.

Measuring circuit: terminal 43 of harness plug FB10.

Standard value: ≈ 0V / low level

Standard value: 10~14V/ high level

Standard value: ≈ 33.2Hz (1000rmp)

Standard value: ≈ 65.4Hz (2000rmp)

Standard value: ≈ 139.7Hz (4200rmp)

7 Check whether the result is normal?

No → Replace engine control module.

Next ↓

Step 6	Complete the check and confirm whether the system works normally nor not.